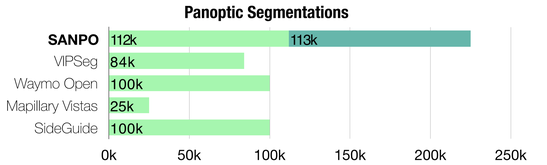
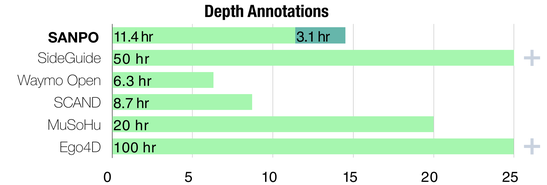
SANPO: A Scene Understanding, Accessibility, and Human Navigation Dataset

Sagar Waghmare, Kimberly Wilber, Dave Hawkey, Xuan Yang, Matthew Wilson, Cattalyya Nuengsigkapian
Stephanie Debats, Astuti Sharma, Lars Pandikow, Huisheng Wang, Hartwig Adam, Mikhail Sirotenko

Comparison to other datasets

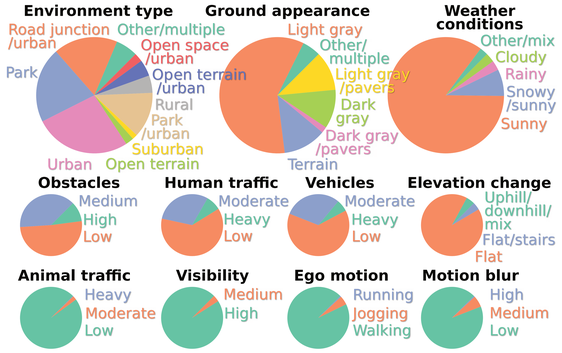




Video dataset comparisons	Panoptic Segmentation	Stereo	Depth	Camera pose	Real + Synthetic	Human Egocentric
SANPO	★	★	★	★	★	★
SideGuide	☆	★	★	-	-	★
Waymo Open	★	★	☆	★	-	-
SCAND	-	☆	☆	★	-	-
MuSoHu	-	★	☆	★	-	★
WVD	-	★	★	★	-	★
Mapillary Vistas	★	-	-	★	-	-
VIPSeg	★	-	-	-	-	-
Ego4D	-	★	★	★	-	★

Data Composition

SANPO-Real contains a variety of outdoor environments and running conditions collected from California, Colorado, and New York City



Assistive technology trained on SANPO helps blind runners run independently.

Vision loss impacts many lives. Globally, 295 million people have moderate to severe visual impairment, and 43 million are blind.

Is there a way to use technology to enable runners with visual impairments to run by themselves? In Japan, around 60 percent of the population exercises routinely. However, that number drops to around 30 percent among people with visual impairments.

Project Guideline, an early-stage research collaboration between Google and Guiding Eyes for the Blind, gives feedback to help blind runners. Leveraging on-device machine learning that runs on Android smartphones, it identifies a colored line on the ground, determines whether the line is to the left, right, or center of the runner, and sends audio signals through headphones worn by the runner. Based on that feedback, the runner can correct their running position to stay on the line and enjoy their run.

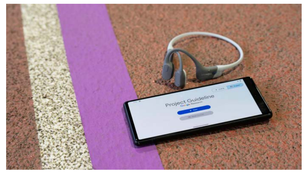
Using only a phone running Guideline technology and a pair of headphones, Guiding Eyes for the Blind CEO Thomas Panek was able to run independently for the first time in decades and complete an unassisted 5K in New York City's Central Park.

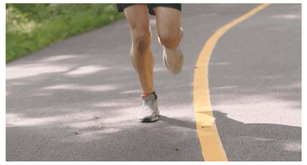
Learn about Project Guideline

Case study: Stockholm Half-Marathon

Initiatives like Project Guideline are a promising start, but there's a long way to go.

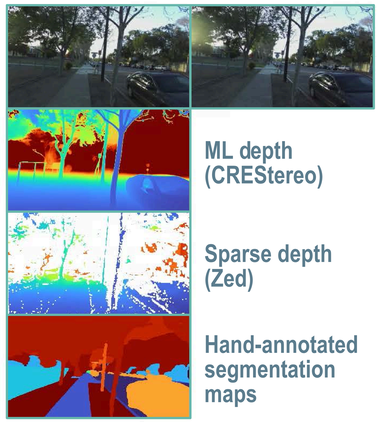
To build better navigation systems, **we need new datasets** that push the frontiers in egocentric outdoor scene understanding. That means datasets that stress-test **monocular depth estimation** and **panoptic video segmentation.** That's where SANPO comes in.





What's included in SANPO?

Head Camera



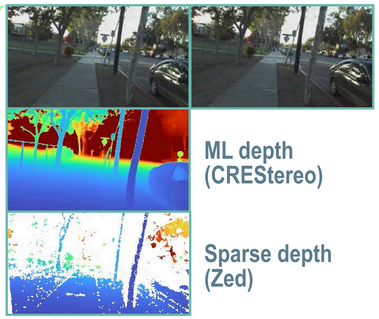
Stereo RGB

ML depth (CREStereo)

Sparse depth (Zed)

Hand-annotated segmentation maps

Chest Camera



Stereo RGB

ML depth (CREStereo)

Sparse depth (Zed)

PII removal

CREStereo

Crowd Labeling

CREStereo

Crowd Labeling

SANPO Real

Zed 2i camera:
- Stereo RGB
- Metric depth
- Segmentation maps∗
- Instance labels∗

∗ : not all sessions

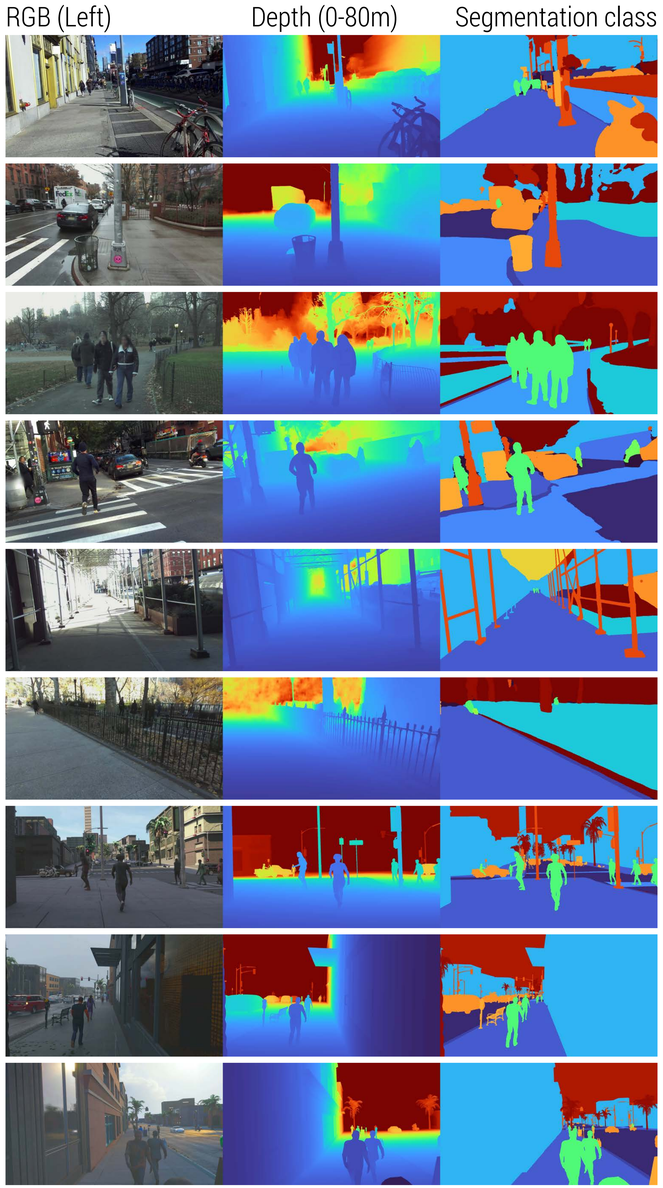
Zed M camera:
- Stereo RGB
- Metric depth
- Segmentation maps∗
- Instance labels∗

∗ : not all sessions

Capture and processing

SANPO-Real was collected with an in-house capture rig worn by volunteer runners. Multiview video is provided by two stereo cameras -- that's 4 RGB streams in total! -- with sparse (Zed SDK) and dense (CREStereo) depth maps. Overall, there are 617,408 frames and depth maps (ignoring stereo). Panoptic video masks are hand-labeled by crowd workers at a reduced frame rate. These masks are machine-propagated to unlabeled frames.

SANPO-Synthetic is a collaboration between Google Research and Parallel Domain, including pixel-perfect depth and panoptic segmentation annotation in realistic environments. There are 113,794 frames in SANPO-Synthetic. **(Can you find the three synthetic images in the examples on the right?)**



Google Research

parallel domain